21-752: Algebraic Topology

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Carnegie Mellon University - Spring 2026

This version of the notes was last compiled on March 20, 2026.

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Course Introduction and Overview

For the purposes of intuition, let's try and think about the concept of geometry. Broadly speaking, we can define geometry to be the study of isometries and their invariants - to talk about distance, we only need to be in a metric space. Topology is a 'squishy' notion of geometry, where instead of distance preservation, we talk about *nearness* preservation.

Already we lose some information. Isometries are automatically injective; moreover, the inverse of the corestriction of an isometry to its image is also an isometry. So distance-preservation is 'invertible' in some sense of the word. Nearness-preservation (which most of us would call "continuity"), however, is famously not (guaranteed to be) invertible. Topology is thus 'more interesting'.

A non-exhaustive list of things we might do with (algebraic) topology:

- **Solve equations implicitly:** for example we may have results analogous to the intermediate value theorem (IVT). In the example of IVT analogues, we can see IVT as "working" because of path-connectedness of the unit interval. Say we want an analogous statement for when maps $f : D^2 \to \mathbb{R}^2$ (with D^2 the disk in $\mathbb{R}^2$) have a 0 - in this instance, we may need a "higher" notion of connectivity...

- **Homotopy:** Just as we can talk about 'higher' notions of connectivity, we can also talk about 'higher' notions of *homotopy*. We are already familiar with questions of path homotopies: can I continuously deform one path to another? We can generalise this in the following manner. If there are two homotopies between two paths, we can ask whether there is a way to continuously deform one *homotopy* to the other. We are all adults here and know how to do induction, so I'm sure we all know where this is going.

- **Local-to-Global:** Do local solutions glue together to give global solutions? Algebraic topology gives us tools that track this.

- **Makes precise ideas such as "testing into/out of an object":** for example, say we want to look at sequences in a metric space X, which are just maps from $\mathbb{N}$ to X. Sequences essentially tell us all we would want to know about continuity in a metric space - so here our "test object" is $\mathbb{N}$, and studying the sequences (i.e. the maps from $\mathbb{N}$ to X) can tell us about the metric space. In topological spaces, maybe we want to look at all continuous maps from n-simplices into our space - in this instance we would have a different "test object" for each dimension, and the test object for dimension n will be the n-simplex $\Delta_n = \{x \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n} x_i = 1\}$. We can use tools from homological algebra to study maps from the n-simplex into the space. The properties we deduce using these methods will tell us a great deal about the space. Moreover, n-simplices tell us about n-fold homotopies (ie, 2-simplices tell us about homotopies of paths; 3-simplices tell us about homotopies of homotopies; and so on).

  As a categorical aside, it turns out that we can define a category Δ known as the **simplex category**, whose objects are finite linear orders (labelled $[n] = \{0 < 1 < \cdots < n\}$) and whose morphisms are weakly order-preserving maps (ie, if $x < y$ then $f(x) \leq f(y)$). We can show that the operation taking any $[n]$ to the n-simplex Δ_n defines a contravariant functor $K : \Delta \to \mathbf{Set}$. Spaces and categories admit a "common generalisation" in simplicial sets; this area is called infinity category theory.

We will spend the first few weeks talking about fundamental groups and how to compute them. We will then move onto 'higher dimensional stuff'.

Chapter 1

The Fundamental Group

Our broad objective will be to *associate invariants to spaces*. Let's begin by putting together a 'wishlist' of things we would want such 'invariants' to obey.

- We want our invariants to be *invariant under deformations*: if we 'do something' to a space ('deform' it), we don't want that to affect our invariant.
- We want our invariants to be *functorial*: an invariant describing spaces should describe groups if we move from spaces to group in a nice way.
- We want our invariants to *completely classify spaces* (possibly up to deformation): in other words, we want a 'converse' to invariance under deformations, where we know that if our invariant is unchanged by some transformation, then that transformation must be a compatible one (a 'deformation').
- We would (ideally) want our definition to be *easily computable*.

Now that we have a list of things we'd like to do, why don't we go ahead and... do them?!

1.1 The Setup

Before we go any further, we will state something (that is par for any course on topology).

Convention. Unless otherwise specified,

- All maps in this course will be assumed to be continuous.
- The (uppercase) letters X, Y and Z will denote topological spaces.

1.1.1 Paths and Homotopies

We begin by defining the notion of 'deformation' we talked about so handwavily just a few moments ago.

Definition 1.1.1 (Homotopy of Continuous Maps)**.** Let X and Y be topological spaces and let $f, g : X \to Y$ be (continuous) functions. A **homotopy from** f **to** g is a (continuous) map $H : X \times I \to Y$ such that for all $x \in X$,

$$H(x, 0) = f(x) \qquad \text{and} \qquad H(x, 1) = g(x)$$

If a homotopy exists between f and g, we say they are **homotopic**, and we write $f \simeq g$.

It is not hard to see that homotopy (or more precisely, the property of being homotopic) is an equivalence relation on the set of continuous functions from one space to another. In particular, it doesn't matter whether we say 'f is homotopic to g' or 'g is homotopic to f'.

Definition 1.1.2 (Paths and Loops)**.** A **path** in a topological space X is a continuous map $\gamma : [0, 1] \to X$. A path γ is a **loop** if $\gamma(0) = \gamma(1)$.

Since life is too short, we will adopt the following notation.

Notation. Unless otherwise specified, the symbol I will denote the unit interval $[0, 1]$.

There are obviously many examples of paths. In fact, there are many examples of homotopic paths.

Example 1.1.3. Any two paths in $\mathbb{R}^n$ (with the same endpoints) are homotopic. For example, if $\alpha, \beta : I \to \mathbb{R}^n$ are paths with the same endpoints, then

$$H : I \times I \to \mathbb{R}^n : (s, t) \mapsto t \cdot \alpha(s) + (1 - t) \cdot \beta(s)$$

is a homotopy between the paths.

Really, it only makes sense to talk about homotopies between paths whose endpoints are the same. That being said, our definition of homotopies is permissive enough that this is not necessarily true of a homotopy. Therefore, we will introduce separate notation for what it means for paths with the same endpoints to be homotopic (notation that emphasises that their endpoints are the same).

Notation. If α and β with the same endpoints, we will write

$$\alpha \simeq_{\partial I} \beta$$

to indicate that α and β are homotopic.

It turns out that paths can be composed, provided they are 'compatible'.

Proposition 1.1.4 (Composing Paths). *Fix points $x_0, x_1, x_2 \in X$. If γ is a path from x_0 to x_1 and δ is a path from x_1 to x_2, then the function*

$$\varepsilon : [0,1] \to X : t \mapsto \begin{cases} \gamma(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ \delta(2t-1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

is a path between x_0 and x_2.

Proof. It is clear that ε is a well-defined function: at the point $t = \frac{1}{2}$,

$$\gamma(2t) = \gamma(1) = x_1 = \delta(0) = \delta(2t-1)$$

So indeed the function is well-defined.

All we need to show is that ε is continuous. But this is true because of a homework exercise from General Topology in which we showed that if we have a space that is a union of two closed sets then we can continuous 'glue' together continuous functions that agree on the overlap of these closed sets. So ε is the 'gluing' of γ of δ, and is hence continuous. □

Add ref

Definition 1.1.5 (Composition of Paths)**.** Fix points $x_0, x_1, x_2 \in X$. If γ is a path from x_0 to x_1 and δ is a path from x_1 to x_2, then the function $\varepsilon : [0,1] \to X$ defined as above is called the **composition** of γ and δ.

We will denote it $\delta \star \gamma$, $\delta * \gamma$, $\delta \cdot \gamma$ or simply $\delta\gamma$.

Warning. We emphasise that $\delta\gamma$ means

First do γ, then do δ.

In other words, we obey the convention we normally use when we do function composition.

Warning. You know everything I just said? Be warned, I may not always heed my own warning and might write $\gamma\delta$ to mean $\delta\gamma$ sometimes. The real warning is, just use your head, typecheck that function compositions and path concatenations are all ok, and you should be fine. Just read the vibe and you should be ok.

Finally, we define the notion of an inverse path.

Definition 1.1.6 (Inverse Path)**.** Given a path $\alpha : I \to X$, we define the **inverse path** $\overline{\alpha} : I \to X$ by $\overline{\alpha}(t) := \alpha(1-t)$ for all $t \in I$.

We will now take a brief detour into the world of groupoids.

1.1.2 Groupoids

Here, we will assume some familiarity with Category Theory. For a quick and dirty introduction to the subject, read `sorry`.

Add refs

Recall that we can view a group as a (small) category, with the morphisms being its action on itself.

Example 1.1.7 (A Group as a (Small) Category)**.** Fix a group G. We can turn it into a small category with one single object $\star$ and one morphism from $\star$ to itself corresponding to each group element g.

We define a groupoid to be a generalisation of the above example that allows for more than one object.

Definition 1.1.8 (Groupoid)**.** A **groupoid** is a small category in which every morphism is invertible.

Comparing Definition 1.1.8 and Example 1.1.7, the following is evident.

Proposition 1.1.9. *A group is a groupoid with only one object.*

1.1.3 The Fundamental Group(oid)

Fix a topological space X. We ask ourselves the following question: can we associate a groupoid to X? As it turns out, we can.

Definition 1.1.10. The **fundamental groupoid of** X, denoted $\pi_1(X)$, is the category with the following data:

- The object set is X.
- For each $x, y \in X$, define the set of morphisms $\mathrm{Hom}(x, y)$ to be the set of all paths from x to y, up to homotopy.
- For $x, y, z \in X$, $[\gamma] \in \mathrm{Hom}(x, y)$ and $[\delta] \in \mathrm{Hom}(y, z)$, define $[\delta] \circ [\gamma]$ to be $[\delta \star \gamma]$.
- For $x, y \in X$ and $[\gamma] \in \mathrm{Hom}(x, y)$, define the inverse operation $[\gamma]^{-1}$ to be $[\overline{\gamma}]$.

We have not really shown that this composition is associative (and of course it is not something that should be shown *after* the definition but rather something that should be shown *before*) but we'll do it now.

Lemma 1.1.11. *Let $\phi : I \to I$ be a path such that $\phi(0) = 0$ and $\phi(1) = 1$. For any path $\gamma : I \to X$, $\gamma \circ \phi \simeq_{\partial I} \gamma$.*

Proof. The function

$$H : I \times I \to X : (s, t) \mapsto \gamma(st + (1 - t)\phi(s))$$

is an explicit homotopy. □

Corollary 1.1.12. *For all paths $\alpha, \beta, \gamma : I \to X$ that are compatible,*

$$\gamma \star (\beta \star \alpha) \simeq_{\partial I} (\gamma \star \beta) \star \alpha$$

Thus, concatenation gives rise to a well-defined (associative) composition on homotopy classes, thus, on the fundamental groupoid.

Recall that a group is a groupoid with a single object. So, it stands to reason that the fundamental groupoid, except when we look at just one point in our space (and paths from that point to itself) rather than *all* points in our space (and *all* paths), should be a group. This is precisely the fundamental group.

Definition 1.1.13 (The Fundamental Group)**.** Fix a point $x_0 \in X$. The **fundamental group of X based at x_0** is the set

$$\pi_1(X; x_0) = \mathrm{Hom}_{\pi_1(X)}(x_0, x_0)$$

Some fundamental groups are easy to compute and some are difficult.

Example 1.1.14. $\pi_1(\mathbb{R}^n)$ is trivial.

1.2 Intuition, Properties and First Examples

In this section, we'll outline important properties of the fundamental group and develop tools to compute fundamental groups.

1.2.1 First Properties

Theorem 1.2.1. *For any space X and basepoint $x_0 \in X$, $\pi_1(X, x_0)$ is a group under the composition operation $[\gamma] \cdot [\gamma'] = [\gamma \star \gamma']$.*

Proof. This should just follow trivially from the definition (because we did 'check' that $\pi_1(X)$ is a groupoid), but let's play the game and check the axioms anyway.

- Identity: The identity element is the loop $c_{x_0} : I \to X : t \mapsto x_0$. Indeed, $c_{x_0} \star \gamma \simeq_{\partial I} \gamma \simeq_{\partial I} \gamma \star c_{x_0}$ via

$$\Phi : I \to I : t \mapsto \begin{cases} 2t & \text{if } t \leq \frac{1}{2} \\ 1 & \text{if } t \geq \frac{1}{2} \end{cases}$$

- Inverses: Recall that we defined the inverse of a path γ to be the path with the time parameter going in the opposite direction - $\overline{\gamma}(t) := \gamma(1-t)$. It's not hard to show that for all loops γ based at x_0, $\gamma \star \overline{\gamma} \simeq_{\partial I} \overline{\gamma} \star \gamma$.

- Associativity: This is precisely Corollary 1.1.12.

□

Theorem 1.2.2. *For X a topological space with $x_0, x_1 \in X$, and $\alpha : I \to X$ a path from x_0 to x_1 ($\alpha(0) = x_0, \alpha(1) = x_1$), then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.*

There is really only one reasonable way to attempt to do this. And when you do it, surprise surprise, it works...

Proof. The desired isomorphism is given by conjugation by the path from x_0 to x_1. Explicitly, if $\alpha : I \to X$ is a path from x_0 to x_1, we can define $\Phi : \pi_1(X, x_0) \to \pi_1(X, x_1)$ by

$$\Phi([\gamma]) := [\overline{\alpha} \star \gamma \star \alpha]$$

It's not very hard to show that this is a group homomorphism: given paths $[\gamma], [\gamma'] \in \pi_1(X, x_0)$, we see that

$$\begin{aligned} \Phi([\gamma] \circ [\gamma']) &= [\overline{\alpha} \star \gamma \star \gamma' \star \alpha] \\ &= [\overline{\alpha} \star \gamma \star \alpha \star \overline{\alpha} \star \gamma' \star \alpha] \\ &= [\overline{\alpha} \star \gamma \star \alpha] \circ [\overline{\alpha} \star \gamma' \star \alpha] \\ &= \Phi([\gamma]) \circ \Phi([\gamma']) \end{aligned}$$

Finally, we can define $\Phi^{-1} : \pi_1(X, x_1) \to \pi_1(X, x_0)$ by

$$\Phi^{-1}([\gamma]) := [\alpha \star \gamma \star \overline{\alpha}]$$

It's not hard to show this is a two-sided inverse. □

Finally, we show that π_1 is functorial.

Proposition 1.2.3. *Let X and Y be topological spaces. Fix a basepoint $x_0 \in X$. For any continuous function $f : X \to Y$, the map*

$$f_* : \pi_1(X, x_0) \to \pi_1(Y, f(x_0)) : \gamma \mapsto [f \circ \gamma]$$

is a well-defined homomorphism of groups.

Proof. sorry

□

1.2.2 Homotopy Invariance

In our initial discussion, we said we wanted the fundamental group to be invariant under certain kinds of deformations of spaces. We begin by defining these deformations and

Definition 1.2.4 (Homotopy Equivalence of Spaces)**.** Let X and Y be topological spaces. If there are maps $f : X \to Y$ and $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$, then we say X and Y are **homotopy equivalent**. We write $X \simeq Y$.

There are many well-known examples of homotopy equivalences of spaces. At some point, we will probably look at **contractible spaces**, which are spaces that are homotopic to a point.

Example 1.2.5 (Contractibility of the Unit Interval)**.** $I \simeq \{0\}$ via the maps

$$f : I \to \{0\} : t \mapsto 0 \qquad g : \{0\} \to I : 0 \mapsto 0$$

Obviously, $f \circ g = \mathrm{id}_{\{0\}}$. We only need a homotopy in the other direction. Indeed, it's not

hard to show that

$$H : I \times I \to I : (s, t) \mapsto st$$

gives the homotopy $g \circ f \simeq \mathrm{id}_I$.

We can see the above example as indicative of a more general pattern - if we have spaces X and Y with $Y \subseteq X$, then it seems that X and Y should be homotopic if there are no "holes" in $X \setminus Y$.

Definition 1.2.6 (Deformation Retraction)**.** Let X be a topological space. Fix $A \subseteq X$ and let $i : A \hookrightarrow X$ denote the inclusion. A map $r : X \to A$ is a **deformation retraction** if $r \circ i = \mathrm{id}_A$ and $i \circ r \simeq \mathrm{id}_X$.

See also [**EPFLTopologie**].

A special property of deformation retractions is that they induce inclusions of fundamental groups.

Convention. Going forward, unless otherwise specified, arbitrary spaces X, Y, Z etc will be assumed to be path-connected (hopefully for obvious and unproblematic reasons). We will also often omit the basepoint if we know we are working in a path-connected space.

Lemma 1.2.7. *Let X and Y be (path-connected) spaces and let $f_0, f_1 : X \to Y$ be homotopic. Then, $(f_0)_* = (f_1)_*$.*

Proof. Let $H : X \times I \to Y$ be a homotopy between f_0 and f_1. The idea is to show that for all $\gamma \in \pi_1(X)$, the path $f_1 \circ \gamma$ is homotopic to the path $f_0 \circ \gamma$. `sorry` □

Theorem 1.2.8. *If we have $X \simeq Y$ then $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$.*

Proof. Assume we have (continuous) functions $f : X \to Y$ and $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We know show that $f_* : \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$ is a homomorphism; in fact, we will show it is an *iso*morphism. It may not be true that $y_0 = f(x_0)$, but since Y is path-connected, we know we have an isomorphism between $\pi_1(Y, f(x_0))$ and $\pi_1(Y, y_0)$.

sorry □

One more useful definition.

1.2.3 The Fundamental Group of the Circle

It is well-known that the fundamental group of the circle (based at any point) is isomorphic to $\mathbb{Z}$. How might one see this?

Let's begin by parametrising the circle in a natural way. Define $p : \mathbb{R} \to S^1 : x \mapsto (\cos(2\pi x), \sin(2\pi x))$. So when we restrict p to I, we go around the circle exactly once.

p is not invertible. In fact, p is periodic, so any point actually has infinitely many pre-images. But p is *locally invertible*: p is invertible on $[0, 1)$, for instance, and all subsets and translates of such subsets.

Define, for all $n \in \mathbb{Z}$, the map $\tilde{\gamma}_n : I \to \mathbb{R} : x \mapsto nx$.

Lemma 1.2.9. *Let $\gamma : I \to S^1$ be a loop based at $x_0 = (1, 0) \in S^1$. Let $\tilde{\gamma} : I \to \mathbb{R}$ be the angle of γ, in the sense that $\tilde{\gamma}(0) = 0$, $\tilde{\gamma}$ is continuous, and $\gamma = p \circ \tilde{\gamma}$. Then, $\gamma \simeq_{\partial I} \tilde{\gamma}_n$, where $n = \tilde{\gamma}(1)$. [Teresa correction: conclusion should be "then $\gamma^n \simeq_{\partial I} p \circ \tilde{\gamma}_n$."]*

Proof. sorry □

Next, we recall Brouwer's Fixed Point Theorem.

Theorem 1.2.10 (Brouwer's Fixed Point Theorem, BFPT)**.** *Any continuous function from the closed unit disc to itself has a fixed point.*

Lemma 1.2.11. *$\pi_1(S^1)$ is nontrivial.*

Proof. Suppose not. Then, for all (continuous) $\gamma : I \to S^1$, there is a homotopy $H : I \times I \to S^1$ with $H(x, 0) = \gamma(x)$ and $H(x, 1) = (1, 0)$. That is, γ is homotopic to the constant path centred at $(1, 0)$. Given that such an H exists, we can find a homotopy $\hat{H} : S^1 \times I \to S^1$, which takes

S^1 to $H(x,1) = (1,0)$ - also, assume the result of Brouwer's FPT (which can be proved without reference to the fundamental group). Then quotienting $S^1 \times I$ by $S^1 \times \{1\}$, we will actually obtain B^2 - but this is a continuous map from B^2 to itself with many fixed points (contradicting the result of Brouwer's FPT) ??? □

1.2.4 The Fundamental Group of the Real Projective Plane

We begin by defining the real projective plane.

Definition 1.2.12 (Real Projective Plane)**.** The **Real Projective Plane**, denoted $\mathbb{R}P^2$, is defined to be the quotient space

$$S^2 \big/ x \sim -x$$

where $x \sim -x$ denotes the **antipodal relation** on S^2.

One can easily show the antipodal relation to be an equivalence relation.

We can visualise $\mathbb{R}P^2$ to be a disc whose boundary is divided into two segments that are identified antipodally: `sorry`

Insert picture

Intuitively, we can think about what loops look like in this space.

There's really only one thing to think about: is a loop homotopically trivial or not? If a loop does not go from one pole to another, it is; if it does cross opposite poles, then it is not.

Lemma 1.2.13. *$\pi_1(S^2)$ is trivial.*

Proof. For any point $y \in S^2$, one can show that $S^2 \setminus \{y\}$ is homeomorphic to $\mathbb{R}^2$ via stereographic projection. So any loop that misses y is homotopic to a point. Thus, as long as we know that every loop misses *some* point, we know that that loop is homotopic to a point.

Now suppose, for contradiction, that there is some loop γ in S^2 that is *surjective*—ie, it misses *no points*. We show that γ is homotopic, with fixed endpoints, to a loop α that agrees with γ everywhere except for an ε-ball around x_0, and α does not map to $B(-x_0, \varepsilon)$.

sorry □

1.3 Covering Spaces

In this section, we outline the basics of covering space theory and explore ways in which covering spaces can help us compute and understand fundamental groups.

Let's begin by appealing to some intuition from our proof that the fundamental group of the circle. A key idea was lifting a loop in the circle into a path from the real line to the circle. We observed that the condition of being homotopically trivial corresponded precisely to the condition of the endpoints of the path in $\mathbb{R}$ lying in the same $\mathbb{Z}$-translate of the fundamental domain. This intuition can be developed and formalised into a much more comprehensive theory.

1.3.1 Definition and First Examples

Throughout this chapter, let X be a topological space.

Definition 1.3.1 (Covering (Space/Map)). A space $\tilde{X}$ along with a map $p : \tilde{X} \to X$ is called a **covering space of** X if the following are all true:

1. p is surjective
2. There is an open covering $\mathcal{U}$ of X such that for all $U \in \mathcal{U}$, there exist open sets $V_\alpha \subseteq \tilde{X}$ such that
   (a) The V_α are all disjoint
   (b) $p^{-1}[U] = \bigcup_\alpha V_\alpha$
   (c) $p|_{V_\alpha} : V_\alpha \to U$ is a homeomorphism

We refer to $\tilde{X}$ as a **covering space** and p as a **covering map**.

First, a concrete example.

Example 1.3.2 ($\mathbb{R}$ covers S^1). $\mathbb{R}$, along with the map $p : \mathbb{R} \to S^1 : t \mapsto e^{2\pi i t}$, is a covering space of S^1.

Next, a handwavy example.

Example 1.3.3 (The Torus covers the Torus). You can 'unravel' the torus into 'two tori stuck together' to form a covering space of the torus.

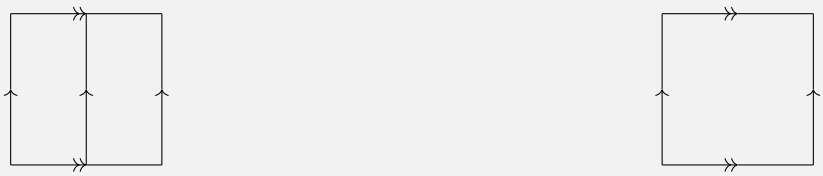


Figure 1.1: A depiction of what's going on

We will see that if a space has a nontrivial fundamental group, then it is the quotient of some space by its fundamental group. Moreover, we will see that this action is "properly discontinuous" and that the parent space, along with the quotient map, form a covering space. So the notion of fundamental groups is fundamentally linked to the notion of covering spaces.

As a side note, we define simply connected spaces.

Definition 1.3.4 (Simply Connected Space). We say that a path-connected space X is **simply connected** if $\pi_1(X) = 0$.

What our above remark tells us about simply connected spaces is that they admit no covering spaces that are acted on properly discontinuously by a (nontrivial) group.

1.3.2 Lifting Theorems

In this subsection, we give two types of (path-)lifting theorems. The first lifts paths (between points) and the second lifts homotopies (paths between paths).

Theorem 1.3.5 (Path Lifting Theorem). *Let $p : \tilde{X} \to X$ be a covering of X. Pick a basepoint $\tilde{x}_0 \in \tilde{X}_0$ and $x_0 \in X$ so that $p(\tilde{x}_0) = x_0$ (i.e. having fixed a basepoint $x_0 \in X$, we choose any $\tilde{x}_0 \in p^{-1}(x_0)$ as a basepoint of $\tilde{X}$). Let $\gamma : I \to X$ be a path, with $\gamma(0) = x_0$.*

There is a unique path $\tilde{\gamma} : I \to \tilde{X}$ such that $p \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(0) = \tilde{x}_0$.

$$\begin{tikzcd} & \tilde{X} \arrow[d, "p"] \\ I \arrow[ur, "\tilde{\gamma}"] \arrow[r, "\gamma"'] & X \end{tikzcd}$$

Proof (see also `sorry`*).* Let $\mathcal{U}$ be a family of evenly covered sets in X. Let $\mathcal{V} = \{\gamma^{-1}[U] \mid U \in \mathcal{U}\}$. This is an open cover of I. Since I is compact, this admits a finite subcover. Thus[1], there are $0 = t_0 < t_1 < \cdots < t_n = 1$ with $\gamma([t_i, t_{i+1}]) \subseteq U$ for some $U \in \mathcal{U}$.

Inductively, if $\tilde{\gamma}$ has been defined on $[0, t_i]$, we can extend $\tilde{\gamma}$ by $(p|_{\tilde{U}})^{-1} \circ \gamma|_{[t_i, t_{i+1}]}$, where $\tilde{U}$ is the connected component of $p^{-1}[U]$ containing $\tilde{\gamma}(t_i)$.

Uniqueness is "by intuition" (i.e. the path is uniquely defined on each set of $\mathcal{V}$, and then since $\mathcal{V}$ covers I uniqueness of $\tilde{\gamma}$ should follow with no additional trouble). □

We can now do the same for paths between paths.

Theorem 1.3.6 (Homotopy Lifting Theorem). *Let $p : \tilde{X} \to X$ be a covering of X. Pick a basepoint $\tilde{x}_0 \in \tilde{x}$ and $x_0 \in X$ so that $p(\tilde{x}_0) = x_0$. Let $F : I \times I \to X$ be a homotopy of paths starting at $x_0 \in X$. There is a unique homotopy $\tilde{F} : I \times I \to \tilde{X}$ lifting F (that is, $p \circ \tilde{F} = F$) with $\tilde{F}(0, 0) = \tilde{x}_0$.*

The paths between which F is a homotopy are precisely

$$\gamma : I \to X : t \mapsto F(t, 0)\delta : I \to X : t \mapsto F(t, 1)$$

and the paths between which $\tilde{F}$ is a homotopy are the unique lifts $\tilde{\gamma}, \tilde{\delta} : I \to \tilde{X}$ given by the path-lifting theorem.

Proof sketch. Let $\mathcal{U}$ be an open cover of X. Split $I \times I$ into finitely many rectangles R_{ij} such that

[1] If I have *any* finite open cover of the unit interval, I can find finitely many such t_i, and I can choose them so that each interval $[t_i, t_{i+1}]$ is contained in an element of that open cover - just draw a picture.

$F[R_{ij}] \subseteq U$ for some $U \in \mathcal{U}$. Construct $\tilde{F}$ by inverting p on the corresponding U and composing with $\tilde{F}$. □

Given all of this, we can now prove that indeed $\pi_1(S^1) = \mathbb{Z}$ in a simpler/cleaner (if equivalent) way.

Theorem 1.3.7. $\pi_1(S^1) = \mathbb{Z}$

Proof. Having already shown (earlier in these notes) that for γ such that $[\gamma] \in \pi_1(S^1)$, letting $\gamma_n : I \to S^1$ such that $\gamma_n : x \mapsto e^{2\pi i n x}$ (for some $n \in \mathbb{Z}$) then we must have $[\gamma] = [\gamma_n]$ for some n, we let $H : I \times I \to S^1$ be a homotopy from γ_n to γ_0. We can now lift H to some $\tilde{H} : I \times I \to \mathbb{R}$, and we'll have $\tilde{H}(\cdot, 0)$ will be a lift of γ_n to $\tilde{H}(1,0) = 0$, $\tilde{H}(1,1) = n$...sorry □

1.3.3 Fundamental Groups of Covering Spaces

There is an immediate (and very strong) statement we can make about the fundamental group of a covering space.

Theorem 1.3.8. *If $p : \tilde{X} \to X$ is a covering with $p(\tilde{x_0}) = x_0$, then the induced map $p_\# : \pi_1\left(\tilde{X}, \tilde{x_0}\right) \to \pi_1(X, x_0)$ is injective.*

Proof. sorry(should follow because path-lifting is unique) □

An immediate consequence is that if a space has trivial fundamental group, so must any covering space of it. That is,

Corollary 1.3.9. *If X is simply-connected, then every covering space of X is also simply-connected.*

But in fact, we can say something even stronger.

Definition 1.3.10 (One-Sheetedness of a Covering)**.** We say that a covering $p : \tilde{X} \to X$ is one-sheeted if $p^{-1}(x)$ has cardinality 1 for all $x \in X$.

Proposition 1.3.11. *Let X be a simply-connected space (and, in particular, path-connected) space and let $p : \tilde{X} \to X$ be a covering. Then, p is one-sheeted.*

Proof. Suppose not. Fix some $x_0 \in X$ and suppose that $p^{-1}(x_0)$ has two distinct points $\tilde{x_1}$ and $\tilde{x_2}$. Let $\tilde{\gamma} : I \to \tilde{X}$ be a path from $\tilde{x_1}$ to $\tilde{x_2}$. Let $\gamma = p \circ \tilde{\gamma}$. This is a loop at x_0. Let γ_t be a homotopy from $\gamma_0 = \gamma$ and $\gamma_1 = c_{x_0}$, the constant path at x_0. We know there is a unique lifted homotopy $\tilde{\gamma}_t$. This is a homotopy from $c_{\tilde{x_1}}$ (and also from $c_{\tilde{x_2}}$) to $\tilde{\gamma}$ with fixed endpoints, so $\tilde{\gamma}$ has to be a loop, meaning $\tilde{x_1} = \tilde{x_2}$. □

In particular, a simply-connected space X can have no non-trivial path-connected covering space.

Our next goal will be to show that path-connected covering spaces of X, up to an appropriate notion of equivalence, are in bijection with subgroups of the fundamental group. See also [**EPFLTopologie**].

Theorem 1.3.12. *Let $p : \tilde{X} \to X$ be a covering, with $p(\tilde{x_0}) = x_0$. Suppose $\tilde{X}$ and X are path-connected. Then the number of sheets of p is equal to the index of $p_\#\left[\pi_1\left(\tilde{X}, \tilde{x_0}\right)\right]$ in $\pi_1(X, x_0)$.*

Proof. For the purposes of this proof, denote

$$G = \pi_1(X, x_0) \qquad\qquad H = p_\#\left[\pi_1\left(\tilde{X}, \tilde{x_0}\right)\right]$$

We can define a map from the set of right-cosets of H to the fibre of x_0 by

$$\phi : \{Hg \mid g \in G\} \to p^{-1}(x_0) : H[\gamma] \mapsto \tilde{\gamma}(1)$$

We need to show that ϕ is well-defined.

- <u>Well-definedness in terms of homotopy class:</u> If γ and δ are such that $[\gamma] = [\delta] \in G$, then we have a unique lifted homotopy from (the lifted paths) $\tilde{\gamma}$ to $\tilde{\delta}$, so they will map to the same element of the fibre (i.e. the lifted paths must end in the same place, as these paths are really loops γ, δ with $[\gamma] = [\delta] \in \pi_1(X, x_0)$). `sorry`

- Well-definedness in terms of cosets: If γ and δ are non-homotopic paths such that $H[\gamma] = H[\delta]$, then $\tilde{\gamma}(1) = \tilde{\delta}(1)$ because for all $[h] \in H$, $\tilde{h}$ is a loop in $\tilde{X}$ (i.e. $\gamma(1) = \gamma(0) = \delta(1) = \delta(0) = x_0$. Then, because there is a unique lift, if the cosets $H[\gamma]$ and $H[\delta]$ are equal, we must have the translates of the constant path at x_0 by γ, δ equal, from which it follows that the lifted paths have the same start/end point). `sorry`

We now claim that ϕ is, in fact, a bijection.

- ϕ is surjective: Fix $\tilde{x_1} \in p^{-1}(x_0)$. Let $\tilde{\gamma}$ be a path from $\tilde{x_0}$ to $\tilde{x_1}$. We know $\gamma = p \circ \tilde{\gamma}$ is a loop based at x_0. In fact,

$$\phi(H[\gamma]) = \tilde{\gamma}(1) = \tilde{x_1}$$

- ϕ is injective: Suppose that $\phi(H[\gamma]) = \phi(H[\gamma'])$ - this would imply that $\tilde{\gamma}(1) = \tilde{\gamma'}(1) \implies \gamma, \gamma'$ translate the constant path to the same point ($\tilde{\gamma}(1) = \tilde{\gamma'}(1)$), which implies that the cosets $H[\gamma], H[\gamma']$ must be the same.

□

Definition 1.3.13 (Semi-Locally Simply-Connected (SLSC)). A space X is **semi-locally simply-connected (SLSC)** if for all $x \in X$, there is some open neighbourhood $U \ni x$ such that the homomorphism $i_\# : \pi_1(U, x) \to \pi_1(X, x)$ induced by the inclusion $i : U \hookrightarrow X$ is the trivial map.

1.3.4 Deck Transformations and Universal Covers

Let X be a topological space and let $p : \tilde{X} \to X$ be a covering. We define a notion of 'automorphisms of covering spaces'.

Definition 1.3.14 (Deck Transformation). A **deck transformation** is an automorophsim of $p : \tilde{X} \to X$, that is, a homeomorphism $f : \tilde{X} \to \tilde{X}$ such that $p \circ f = p$, ie, the following

diagram commutes:

$$\begin{tikzcd} \tilde{X} \arrow[rr, "f"] \arrow[dr, "p"'] & & \tilde{X} \arrow[dl, "p"] \\ & X & \end{tikzcd}$$

The set of deck transformations forms a group $\mathrm{Deck}(p)$ under composition.

We can show the following useful fact.

Proposition 1.3.15. *If $\tilde{X}$ is simply connected then* $\mathrm{Deck}(p) \cong \pi_1(X)$.

Now that we have a definition of automorphisms of covering spaces, we can define special objects via some universal property.

Theorem 1.3.16 (Existence of Universal Covers). *Let X be*

- *path-connected*
- *locally path-connected*
- *SLSC*

Then there is a universal cover $p : \tilde{X} \to X$ of X.

Proof. Pick a basepoint $x_0 \in X$. Define $\tilde{X}$ to be the following set of homotopy classes (with fixed endpoints):

$$\tilde{X} = \{[\alpha] \mid \alpha : I \to X \text{ is a path with } \alpha(0) = x_0\}$$

Define a covering map

$$p : \tilde{X} \to X : [\alpha] \mapsto \alpha(1)$$

This is well-defined (ie, doesn't depend on the homotopy class) because the homotopy classes are taken with fixed endpoints.

sorry □

Lemma 1.3.17. *Let $p : \tilde{X} \to X$ be a covering space with $p(\tilde{x_0}) = x_0$ and let $f : Y \to X$ with $f(y_0) = x_0$ with Y a space which is path-connected and locally path-connected. Then a lift $\tilde{f} : Y \to \tilde{X}$ of f exists is equivalent to the statement that $f_\#(\pi_1(Y, y_0)) \subseteq p_\#(\pi_1(\tilde{X}, \tilde{x_0}))$.*

Proof. for the forwards direction, $p \circ \tilde{f} = f \implies p_\# \circ f_\# = f_\# \implies \mathrm{im}(f_\# \subseteq \mathrm{im}(p_\#)$.

for the other, we construct $\tilde{f}$. Let $y \in Y$ be arbitrary and let γ be a path from y_0 to y. then $f \circ \gamma$ is a path from x_0 to $f(y)$. Lift this path $(f \tilde{\circ} \gamma) : I \to \tilde{X}$, then we have $\tilde{f}(y) = (f \tilde{\circ} \gamma)(1)$. We want this to be well-defined... □

Lemma 1.3.18. *If X is path-connected and locally path-connected with $p_1 : \tilde{X_1} \to X$ and $p_2 : \tilde{X_2} \to X$ a path-connected covering space of X, then p_1, p_2 are isomorphic via $f : \tilde{X_1} \to \tilde{X_2}$ that maps $\tilde{x_1} \in p_1^{-1}(x_0)$ to $\tilde{x_2} \in p_2^{-1}(x_0)$ (or equivalently we will have that $(p_1)_\#(\pi_1(\tilde{X_1}, x_1)) = (p_2)_\#(\pi_1(\tilde{X_2}, \tilde{x_2}))$.)*

Proof. for the forwards direction $p_2 \circ f = p_1 \implies (p_2)_\# \circ f_\# = (p_1)_\#$ and the other way around $\implies \mathrm{im}((p_1)_\#) = \mathrm{im}((p_2)_\#)$...sorry □

To summarize after this last proof (which will one day be completed), X path-connected, locally path-connected and SLSC, then every subgroup $H \leq \pi_1(X, x_0)$ there is **at most** one covering space $p : \tilde{X} \to X$ with $p(\tilde{x_0}) = x_0$ and $H = p_\#(\pi_1(\tilde{X}, \tilde{x_0}))$.

For such X we have constructed a simply connected covering space $p : \tilde{X} \to X$ and $\pi_1(X, x_0)$ acts on $\tilde{X}$ by deck transformations.

Definition 1.3.19 (G-actions on spaces)**.** Let X be a space and G a group, then a "G-action on X" is a homomorphism $G \to \mathrm{Homeo}(X)$.

Notation. We say that $g \in G$ induces the homeomorphism $x \mapsto gx$ and $gh : x \mapsto (gh) \cdot x = g(h \cdot x)$

An example is given if we let $\mathbb{Z}$ act on $\mathbb{R}$ by translation. sorry... [teresa - 20ish minutes of class

were missed, my apologies]

Definition 1.3.20 (Normal Covering Spaces). Let $p : \tilde{X} \to X$ be a covering space. Then we say that p is "normal" if $\forall x \in X$ and $\forall \tilde{x_1}, \tilde{x_2} \in p^{-1}(x)$ there is a deck transformation that maps $\tilde{x_1}$ to $\tilde{x_2}$.

Proposition 1.3.21. *Normal covering spaces correspond to normal subgroups of $\pi_1(X, x_0)$.*

Proof. Let $p : \tilde{X} \to X$ be a covering space with $\tilde{X}$ path-connected and locally path-connected, let $H = p_\#(\pi_1(\tilde{X}, \tilde{x_0}))$. Recall that the "normalizer" of H a subgroup of G is given by $N(H) := \{g \in G : gH = Hg\}$ and also recall that $p : \tilde{X} \to X$ a covering space which is path-connected, then there is a deck transformation $f : \tilde{X} \to \tilde{X}$ that maps $\tilde{x_0} \in p^{-1}(x_0)$ to $\tilde{x_1} \in p^{-1}(x_0)$. Equivalently, $p_\#(\pi_1(\tilde{X}, \tilde{x_0})) = p_\#(\pi_1(\tilde{X}, \tilde{x_1}))$. Now $p_\#(\pi_1(\tilde{X}, \tilde{x_0})) = H$ and $p_\#(\pi_1(\tilde{X}, \tilde{x_1})) = [\gamma]H[\gamma]^{-1}$ where $\gamma = p \circ \alpha$ with $\alpha : I \to \tilde{X}$ a path from $\tilde{x_0}$ to $\tilde{x_1}$. Thus such a deck transformatition exists iff $[\gamma] \in N(H)$...sorry. □

Theorem 1.3.22. *$p : \tilde{X} \to X$ a covering space with $\tilde{X}$ path connected/locally path-connected, then Deck$(p) \cong N(H)/H$ where $H = p_\#(\pi_1(\tilde{X}, \tilde{x_0}))$.*

Proof. Let $\phi : N(H) \to \text{Deck}(p)$ with $[\alpha] \mapsto \tau$ where $\tau : \tilde{X} \to \tilde{X}$ is the deck transformation that maps $\tilde{x_0} = \tilde{\alpha}(0)$ to $\tilde{x_1} = \tilde{\alpha}(1)$. Note that ϕ is a group homomorphism, and $\phi([\gamma][\gamma^{-1}]) = \phi([\gamma]) \cdot \phi([\gamma^{-1}]) = \tau$. Now $\gamma \star \gamma'$ lifts to $\tilde{\gamma} \star \tau \circ \tilde{\gamma'}$ a path from $\tilde{x_0}$ to $\tau(\tilde{x_2}) = \tau \circ \tau'(\tilde{x_0})$, the deck transformation that maps $\tilde{x_0}$ to $\tau \circ \tau'(\tilde{x_0})$ is $\tau \circ \tau'$, so $\phi([\alpha]) = \text{id}$ iff $\tilde{\gamma}$ is a loop iff $[\alpha] \in H \implies \ker(\phi) = H$...note that the intersection of these two parts is not path-connected.sorry □

After having developed some theory about covering spaces, we have some algebraic applications:

Theorem 1.3.23 (Nielsen-Schreier Theorem). *Any subgroup of a free group is free.*

Proof. (sketch) We consider the free group generated by a set S and observe that F_S (the free group on S) is isomorphic to $\pi_1(\vee_S S^1$ (a wedge of S circles). Let $H < F_S$, then H corresponds to some covering space $p : X_H \to X$, let $H = p_\#(\pi_1(X_H))$. Since p is a local homeomorphism,

X_H is a graph (of vertices and edges). Let T be a maximal tree in X_H, then T is contractible and $X_H/T \cong \vee_S S^1$. Now (one can show) $X_H/T \simeq X_H$, so $\pi_1(X_H)$ is free. □

and a classic

Theorem 1.3.24 (Fundamental Theorem of Algebra). *If $p : \mathbb{C} \to \mathbb{C}$ is a nonconstant polynomial, p has a 0.*

Proof. Divide everything by its absolute value to get $\hat{p} : \mathbb{C} \to S^1$, the $\hat{p}$ restricted to $\partial B_r(0)$ (the ball of radius r centered at 0) and to $\partial B_{r-1}(0)$ are homotopic. Now...sorry □

We also claim that if G is any finitely generated group, then there is a space X with $\pi_1(X) \cong G$. To sketch the proof, we want to let $\tilde{X} = G \star G \star G$ the "three-fold join of G" which we build as a tri-partite graph with each part having vertices in G, and then we glue in a triangle for every choice of point (one from each column). This has a natural G-action which is properly discontinuous, and then $X = \tilde{X}/G$...and "this is the start of group cohomology"...

1.4 Applications of the Fundamental Group (and everything else we've done so far)

1.4.1 Non-Homeomorphicity of Euclidean Spaces

The following, while intuitive, is a rather difficult fact to prove: that if $m \neq n$, then $\mathbb{R}^m \ncong \mathbb{R}^n$.

The proof that $\mathbb{R} \ncong \mathbb{R}^2$ (or, for that matter, that $\mathbb{R} \ncong \mathbb{R}^n$ for $n > 1$) is simple: if there were a homeomorphism, then removing a point from $\mathbb{R}$ should give us a homeomorphism to the space obtained by removing the corresponding point from the higher-dimensional $\mathbb{R}^n$, but no such homeomorphism can exist, because removing a point from $\mathbb{R}$ disconnects $\mathbb{R}$ but removing a point from $\mathbb{R}^n$ does not disconnect $\mathbb{R}^n$ if $n > 1$.

We can generalise this proof to show that $\mathbb{R}^2 \ncong \mathbb{R}^n$ for $n > 2$, but we cannot do so using just connectivity; rather, we need higher notions of connectivity.

Theorem 1.4.1. *For all $n \neq 2$, $\mathbb{R}^2 \neq \mathbb{R}^n$.*

Proof. We have outlined the $n = 1$ case already at the start of this subsection, so we will only do the $n > 2$ case.

Suppose $\mathbb{R}^2 \cong \mathbb{R}^n$ via some homeomorphism $f : \mathbb{R}^2 \to \mathbb{R}^n$. Then, $\mathbb{R}^2 \setminus \{0\} \cong \mathbb{R}^n \setminus \{f(0)\}$. Note that $\mathbb{R}^2 \setminus \{0\} \simeq S^1$ and $\mathbb{R}^n \setminus \{f(0)\} \simeq S^{n-1}$. Thus, we have a homotopy equivalence from S^1 to S^{n-1} via

$$S^1 \simeq \mathbb{R}^2 \setminus \{0\} \cong \mathbb{R}^n \setminus \{f(0)\} \simeq S^{n-1}$$

But $\pi_1(S^1) = \mathbb{Z} \ncong 0 = \pi_1(S^{n-1})$. □

1.4.2 Surfaces and Genuses

Theorem 1.4.2. *The closed, orientable surface of genus g is not homeomorphic to the closed, orientable surfact of genus $g' \neq g$.*

Proof sketch. The two are not even homotopy equivalent, because their fundamental groups are different. Indeed, the abelianisation of the fundamental group of the COSG[2] g is $\mathbb{Z}^{2g}$. □

1.4.3 The Borsuk-Ulam Theorem

Lemma 1.4.3. *Let $\gamma : I \to S^1$ be a loop, ie, a continuous map with $\gamma(0) = \gamma(1)$. If*

$$\gamma\left(t + \frac{1}{2}\right) = -\gamma(t)$$

for all $t \in I$, then γ is homotopically non-trivial.

Proof. We know that $p : \mathbb{R} \to S^1$ is a universal cover. Lift γ to a path $\tilde{\gamma} : I \to \mathbb{R}$, with $\tilde{\gamma}(0) = 0$ and $p \circ \tilde{\gamma} = \gamma$. `sorry` □

[2]closed orientable surface of genus

Theorem 1.4.4 (Borsuk-Ulam). *Given any continuous $f : S^2 \to \mathbb{R}^2$, there exists some $x \in S^2$ such that $f(x) = (-x)$.*

Proof. □

There are many applications of the Borsuk-Ulam Theorem—so many, in fact, that you can fill an entire book with them [**Using-Borsuk-Ulam**].

One such example is the non-planarity of K_5, the complete graph on five vertices.

Theorem 1.4.5. *There is no continuous injection from K_5 to $\mathbb{R}^2$.*

Proof sketch. Let $C_2(K_5) = \{(x, y) \in K_5 \times K_5 \mid x \neq y\}$. Suppose $f : K_5 \hookrightarrow \mathbb{R}^2$ is a continuous injection. We get a map

$$F : C_2(K_5) \to S^1 : (x, y) \mapsto \frac{f(x) - f(y)}{\|f(x) - f(y)\|}$$

We can find $h : S^2 \to C_2(K_5)$ such that if $h(u) = (x, y)$ then $h(-u) = (y, x)$. Then, the map

$$S_2 \xrightarrow{h} C_2(K_5) \xrightarrow{F} S^1$$

contradicts the Borsuk-Ulam Theorem. □

1.4.4 Hanging a Rope on Two Nails

We can ask the following question: if I have two nails on a wall, how can I hang a rope on them in such a way that if I remove either one of the nails, the rope falls off?

We can make the following analogy: let the wall be $\mathbb{R}^2$ and let the nails be points in $\mathbb{R}^2$. The rope corresponds to a loop, and falling off corresponds to being homotopically trivial.

We know that $\mathbb{R}^2$ without two points is homotopic to a wedge sum of two circles, and we know this has fundamental group F_2, the free group on two generators. Call the generators a and b. Our question asks if we can find a (nontrivial) element of F_2 such that when you remove all occurrences of either a or b, everything cancels. But this is easy: just take $aba^{-1}b^{-1}$.

1.5 Van Kampen's Theorem

In the spirit of computing fundamental groups (before moving on to higher notions of homotopy), we now consider how/in what way the fundamental group of a space X can be understood in terms of the fundamental groups of different "pieces" of X.

Example 1.5.1. To get an idea of the kind of result we might expect (or could hope to have), we look at a few examples:

- We know that $\pi_1(S_1) \cong \mathbb{Z}$, and $\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z}$ (the free product of $\mathbb{Z}$ with itself). $S^1 \vee S^1$ is just two circles glued together at a point.
- We know that $\pi_1(S^1 \times S^1) \cong \mathbb{Z}^2$...we can imagine $S^1 \times S^1$ as $S^1 \vee S^1$, with a square glued along the boundary - in other words, we can imagine filling in the boundary of $S^1 \vee S^1$ with a disk D, visualized as a square with opposite sides identified. Now $\pi_1(D)$ is trivial, so if we imagine that a, b generate $S^1 \vee S^1$ then we can see that a loop in $S^1 \times S^1$ will be trivial exactly when it is in the kernel of the quotient of our square (i.e. we can see that $\pi_1(S^1 \times S^1) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$). We can think then of $S^1 \times S^1$ as a union of the disk D and the quotiented boundary (isomorphic to $S^1 \vee S^1$) of our disk...and to get the fundamental group, we quotient $\pi_1(S^1 \vee S^1)$ by the (homotopy-class of) trivial loops in the intersection of $D \cap S^1 \vee S^1$...
- We could also consider S^1 composed of "parts" $S^1 = (S^1 \setminus \{u_1\}) \cup (S^1 \setminus \{u_2\})$ for some distinct $u_1, u_2 \in S^1$. Each of these parts $S^1 \setminus \{u_i\}$ has trivial π_1, and we know that $\pi_1(S^1) \cong \mathbb{Z}$. `sorry`

The long and the short of Van Kampen's Theorem is that under the right conditions, π_1 preserves pushouts.

Recall that the free product is the coproduct in the category of groups. This means that if G_α and H are groups, and $\varphi_\alpha : G_\alpha \to H$ are homomorphisms, then there is a unique extension $\phi : *_\alpha G_\alpha \to H$.

In our setup, let X be a space and let $x_0 \in X$ be a basepoint. Assume there exist open, path-connected sets A_α with $x_0 \in \bigcap_\alpha A_\alpha$ and $X = \bigcup_\alpha A_\alpha$.

We know that the inclusions $A_\alpha \hookrightarrow X$ induce maps $j_\alpha : \pi_1(A_\alpha; x_0) \to \pi_1(X; x_0)$. Since $*$ is a

coproduct in **Grp**, these are uniquely extended to a homomorphism $\phi : *_\alpha \pi_1(A_\alpha; x_0) \to \pi_1(X; x_0)$.

Lemma 1.5.2. *ϕ, as defined above, is surjective provided that* `sorry`.

We can summarise everything we've done so far into one gigantic theorem. See also [**EPFLTopologie**].

Theorem 1.5.3 (Van Kampen). *Let X be a topological space with basepoint x_0. Assume there exist sets A_α that such that*

- *Each A_α is open and path-connected*
- $x_0 \in \bigcap_\alpha A_\alpha$
- $A = \bigcup_\alpha A_\alpha$
- *For all α, β, $A_\alpha \cap A_\beta$ are path-connected*

Then, the map $\phi : *_\alpha \pi_1(A_\alpha; x_0) \to \pi_1(X; x_0)$, obtained as a coproduct of the maps induced by the inclusions, is surjective.

If, moreover, $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected for all α, β, γ, then $\ker(\phi)$ is the normal subgroup generated by $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta; x_0)$, where $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta; x_0) \to \pi_1(A_\alpha; x_0)$ is the map induced by the inclusion.

In particular,

$$\pi_1(X; x_0) \cong *_\alpha \pi_1(A_\alpha; x_0) \big/ \ker(\phi)$$

via the first isomorphism theorem.

Remark. Before we prove this theorem, we make a few remarks.

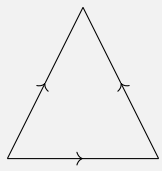
1. Compare the statement above to the statement that if there exist $A, B \subseteq X$ open such that $X = A \cup B$ and $A \cap B$ is path-connected, then $\pi_1(X) = \pi_1(A) *_{\pi_1(A \cap B)} \pi_1(B)$, ie, π_1 preserves pushouts under the right conditions [**EPFLTopologie**]. The quotient group in the version of the theorem stated in lectures is precisely the amalgamated product. Right?

2. Going from a 2 set version to an n set version shouldn't be hugely complicated, as long as we're careful about taking quotients and amalgamated products and whatever

For the proof, we can more or less follow [**EPFLTopologie**].

Proof. The only thing that we really need to check is that $\ker(\phi)$ is generated by the right elements, since we have already shown surjectivity in Lemma 1.5.2. sorry □

Example 1.5.4 (The Dunce Hat)**.** The **dunce hat** is the topological space obtained by making the following identifications on the (solid) triangle:



sorry

Corollary 1.5.5. *Given path-connected spaces* $(X; x_0)$ *and* $(Y; y_0)$*,*

$$\pi_1(X \vee Y) \cong \pi_1(X; x_0) * \pi_1(Y; y_0)$$

where we wedge them at their basepoints.

Example 1.5.6 (Wedge of Projective Planes)**.** Recall that $\pi_1(\mathbb{R}P^2) \cong \mathbb{Z}/2\mathbb{Z}$. Thus,

$$\pi_1(\mathbb{R}P^2 \vee \mathbb{R}P^2) \cong \mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/2\mathbb{Z}$$

This is a group where we have two generators, a and b, and all powers of a and powers of b are trivial. So words look like

$$a, b, ab, ba, aba, bab, abab, baba, ababa, babab, \ldots$$

We can, in fact, use this information to construct a universal cover of $\mathbb{R}P^2$: such a cover consists of countably many copies of S^2, where each is joined to the next at a pole (so it's a row of S^2s where the each is wedged to the next at its north pole/the other's south pole).

Chapter 2

Homology

We motivate the idea of homology by saying a few brief words about higher homotopy groups.

Just as the fundamental group is a group of (pointed) maps from S^1 to a space, up to an appropriate notion of equivalence, higher homotopy groups π_n are groups of (pointed) maps from S^n to a space, up to an appropriate notion of equivalecne ("higher homotopy").

These are, in general, difficult to compute. Consider, for example, $\pi_3(S^2)$. This is generated by the Hopf fibration [**EPFLTopologie**]

$$h : S^3 \to S^2 : (z_1, z_2) \mapsto \frac{z_1}{z_2}$$

where we view S^3 as a subset of $\mathbb{C}^2$ and S^2 as the one-point compactification of $\mathbb{C}$ (the Riemann sphere).

"Since that is annoyingly difficult, let's do something easier - homology."

This theory will be "nice" in several ways - we will lose track of basepoints, and as everything is abelian this will essentially boil down to linear algebra.

2.1 A Fully Abelian Theory

π_1 keeps track of how points and paths interact.

2.1.1 A Boundary Map

Let X be a topological space. Let $P = \{\gamma : I \to X \mid \gamma \text{ is continuous}\}$ be the set of all paths in X. Consider $\mathbb{R}^{\oplus X}$ and $\mathbb{R}^{\oplus P}$, the (free) vector spaces with basis X and P respectively. [Recall/note that $\mathbb{R}^{\oplus X}$ denotes the set/space of all finite linear combinations of elements of X.]

We define a linear map $\partial : \mathbb{R}^{\oplus P} \to \mathbb{R}^{\oplus X}$ on basis elements of $\mathbb{R}^{\oplus P}$ by

$$\partial(\gamma) := \gamma(1) - \gamma(0)$$

We sometimes refer to ∂ as a "boundary map" because it literally involves studying the boundaries of maps.

$\mathrm{im}(\partial)$ keeps track of path components.

Lemma 2.1.1. *The dimension of* $\mathrm{coker}(\partial) = \mathbb{R}^{\oplus X} / \mathrm{im}(\partial)$ *is precisely the number of path-components of* X.

An outline. This is evidently true if X is path-connected: we know that $\mathbb{R}^{\oplus X} / \mathrm{im}(\partial)$ has dimension at least 1 since every vector in $\mathrm{im}(\partial)$ has coordinate 0. It has dimension at most 1 as we can fix $x_0 \in X$ and let $y \in X$ be arbitrary. Then $y - x_0 \in \mathrm{im}(\partial)$ since there is a path connecting y, x_0..."in general, we can write $X = \cup_i X_i$ as a union of its path components, and we can note that ∂ splits along X_i. □

Note that $\mathrm{coker}(\partial)$ is precisely the **zeroth homology of** X (think kernel mod image where the kernel is of the zero map).

Note also that we don't need to take $\mathbb{R}$ to be the base field. You could work in the category of abelian groups and take the coefficients to be in $\mathbb{Z}$ (so you are working with (free) $\mathbb{Z}$-modules instead). For that matter, you could work in the category of modules over any commutative ring.

2.1.2 Higher-Dimensional Simplices

We want to continue what we've been doing to 'higher dimensions'.

Consider

$$T = \{f : \Delta_2 \to X \mid f \text{ is continuous}\}$$

where

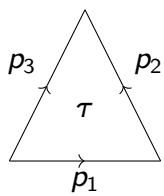
$$\Delta_2 = \left\{x \in \mathbb{R}^2 \;\middle|\; x_i \geq 0,\ x_1 + x_2 + x_3 = 1\right\}$$

Ie, Δ_2 is the 2-simplex, known more commonly as the "triangle".

Consider $\mathbb{R}^{\oplus T}$, the $\mathbb{R}$-vector space with basis T. Define

$$\partial : \mathbb{R}^{\oplus T} \to \mathbb{R}^{\oplus P}$$

defined on the basis of $\mathbb{R}^{\oplus T}$ in the following manner: for all $\tau \in T$, define $\partial(\tau) := p_1 + p_2 - p_3$, where p_1, p_2 and p_3 are the paths in the following picture:



We can continue in this manner with higher-dimensional simplices. Surprise surprise, they form a chain complex: indeed, in the above case

$$\mathbb{R}^{\oplus T} \xrightarrow{\partial} \mathbb{R}^{\oplus P} \xrightarrow{\partial} \mathbb{R}^{\oplus X}$$

we can see that $\partial^2(\tau) = 0$ for all τ, because $p_1 + p_2 - p_3$ is a loop (so its start point minus its end point is zero). We can show something analogous when we consider boundary maps between free modules generated by paths from higher-dimensional simplices as well.

We can thus compute homology as usual:

$$H_n(X; \mathbb{R}) := {}^{\ker(\partial)}\big/_{\mathrm{im}(\partial)}$$

(and similarly for $H_n(X; R)$ where R is any commutative ring).

2.1.3 A Formal Definition (of Singular Homology)

We begin by defining the n-simplex.

Definition 2.1.2 (n-Simplex)**.** For all $n \in \mathbb{N}$, define the n-**simplex** to be the following subset of $\mathbb{R}^n$:

$$\Delta^n := \left\{ x \in \mathbb{R}^n \;\middle|\; \sum_{i=1}^{n} x_i = 1 \text{ and } \forall i,\ x_i \geq 0 \right\}$$

Simplices are well-understood, particularly in low dimensions:

1. The 0-simplex is a point.
2. The 1-simplex is a line.
3. The 2-simplex is a triangle.
4. The 3-simplex is a tetrahedron.

Definition 2.1.3 (Singular n-Simplex)**.** Let X be a topological space. A continuous map $\sigma : \Delta^n \to X$ is called a **singular** n-**simplex**.

For the remainder of this subsection, let G denote an additively expressed abelian group. Unless otherwise specified, G will denote the integers.

Definition 2.1.4 (Group of n-Chains)**.** For all $n \in \mathbb{N}$, define the **group of** n-**chains** $C_n(X; G)$ to be the group whose elements are formal finite linear combinations $\sum_{i=1}^{k} \lambda_i \sigma_i$ of singular n-simplices $\sigma_i : \Delta^n \to X$ with coefficients $\lambda_i \in G$.

Before proceeding any further, we introduce some notation.

Notation. We denote by $[v_0, \ldots, \widehat{v_i}, \ldots, v_n]$ the tuple $[v_0, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n]$.

Now, we define the boundary map.

Definition 2.1.5 (Boundary Map)**.** We let $\partial_n : C_n(X;G) \to C_{n-1}(X;G)$ denote the (n^{th}) **boundary map**, where ∂_n is defined on a basis of $C_n(X;G)$ by

$$\partial_n(\sigma) := \sum_{i=0}^{n} (-1)^i \sigma \restriction_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]}$$

where $[v_0, \ldots, v_n]$ is set of vertices of $\triangle^n$ (i.e. standard basis of $\mathbb{R}^n$).

The kernel of the boundary map consists of all cycles and the image consists of all boundaries.

Lemma 2.1.6. $\partial_n \circ \partial_{n+1} = 0$.

Proof. Fix $\sigma \in C_n(X;G)$. We have

$$\begin{aligned}
\partial_n(\partial_{n+1}(\sigma)) &= \partial_n\left(\sum_{i=0}^{n+1} (-1)^i \cdot \sigma|_{[v_0,\ldots,\widehat{v_i},\ldots,v_{n+1}]}\right) \\
&= \sum_{i=0}^{n+1}\sum_{j<i} (-1)^i (-1)^j \cdot \sigma|_{[v_0,\ldots,\widehat{v_j},\ldots,\widehat{v_i},\ldots,v_{n+1}]} + \sum_{i=0}^{n+1}\sum_{j>i} (-1)^i (-1)^{j-1} \cdot \sigma|_{[v_0,\ldots,\widehat{v_i},\ldots,\widehat{v_j},\ldots,v_{n+1}]} \\
&= \sum_{i=0}^{n+1}\sum_{j<i} (-1)^i (-1)^j \cdot \sigma|_{[v_0,\ldots,\widehat{v_j},\ldots,\widehat{v_i},\ldots,v_{n+1}]} + \sum_{j=0}^{n+1}\sum_{i>j} (-1)^j (-1)^{i-1} \cdot \sigma|_{[v_0,\ldots,\widehat{v_j},\ldots,\widehat{v_i},\ldots,v_{n+1}]} \\
&= 0
\end{aligned}$$

Thus, $\partial_n \circ \partial_{n+1} = 0$. □

Thus, $\mathrm{im}(\partial_n) \subseteq \mathrm{im}(\partial_{n+1})$, and the following is a chain complex in the category of abelian groups:

$$\cdots \longrightarrow C_{n+1}(X;G) \xrightarrow{\partial_{n+1}} C_n(X;G) \xrightarrow{\partial_n} C_{n-1}(X;G) \xrightarrow{\partial_{n-1}} \cdots \tag{2.1.1}$$

Definition 2.1.7 (Cycles)**.** We define a **cycle** in $C_n(X;G)$ to be an element of $\ker(\partial_n)$. We denote $Z_n(X;G) := \ker(\partial_n)$.

Definition 2.1.8 (Boundary Elements)**.** We define a **boundary element** of $C_{n-1}(X;G)$ to be an element of $\mathrm{im}(\partial_n)$. We denote $B_n(X;G) := \mathrm{im}(\partial_n)$.

We are now finally ready to define the nth homology group of topological space.

Definition 2.1.9 (nth Homology Group)**.** We define the **nth homology group of X with respect to G** to be the quotient group

$$H_n(X;G) := \ker(\partial_n)\big/\mathrm{im}(\partial_{n+1}) = Z_n(X;G)\big/B_{n+1}(X;G)$$

We can also define

- **simplicial homology**, where we divide a space into simplices and proceed geometrically. This coincides with singular homology.
- **reduced homology**, which is exactly the same as singular homology except at index 0, where we define the boundary map to go from $C_0(X;G)$ to G in a way that sends every σ to 1.

As we would expect, homology is functorial. As a first step, we show that continuous maps induce morphisms of chain complexes on the chains of cycles. Then, as we know and love, we apply the fact that chain complex morphisms induce homology morphisms.

Moreover, as we would expect, under this correspondence, a homotopy equivalence of topological spaces maps to a homotopy equivalence of chain complexes, which results in an isomorphism on homology.

Finally (as is obvious from functoriality), if we have a homeomorphism, then we get a chain complex isomorphism, and subsequently an isomorphism of homology groups.

2.2 Complex Structures

Let X be a topological space.

Definition 2.2.1 (Δ-complex)**.** A Δ-complex structure on X is a set of singular n-simplices $\delta_\alpha :' \Delta^n \to X$ for varying $n = n(\alpha)$ such that

1. every $x \in X$ is the image of at least one δ_α, $\sigma_\alpha|_{\mathrm{int}(\Delta^n)}$ is injective
2. $\sigma_\alpha|_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]}$ is $\sigma_\beta : \Delta^{k-1} \to X$ in the order-preserving identification of vertices
3. $U \subseteq X$ is open iff $\sigma_\alpha^{-1}(U)$ is open for all α.

This is the "minimal definition" for making the definition of homology work.

Our objective will be to show that homotopies of spaces induce homotopies of chain complexes.

2.2.1 Prisms

Definition 2.2.2 (n-Prism)**.** For all $n \in \mathbb{N}$, define the n-prism to be $\Delta^n \times I \subseteq \mathbb{R}^n$.

This is not a simplex, so we need to triangulate: we can view the n-prism as a(n uncountable) union of $(n-1)$-simplices. Denote the bottom n-simplex of an n-prism by $v_0, \ldots, v_n$ and the top n-simplex by $w_0, \ldots, w_n$.

Definition 2.2.3 (Prism Operator)**.** Define $P : C_n(X) \to C_{n+1}(X)$ by

$$P(\sigma) = \sum_i (-1)^i F \circ (\sigma \times \mathrm{id})|_{[v_0,\ldots,v_i,w_i,\ldots,w_n]}$$

Proposition 2.2.4. *Given $f, g : X \to Y$, if*

$$\partial \circ P = g_\# - f_\# - P \circ \partial$$

then the maps induced by f and g on homology

$$f_*, g_* : H_*(X) \to H_*(Y)$$

are equal.

Definition 2.2.5 (Chain Homotopy)**.** A sequence of maps $P_n : C_n(X) \to C_{n+1}(X)$ is called a **chain homotopy** if

$$\partial \circ P + P \circ \partial = g_\# - f_\#$$

We keep in mind the following commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r, "\partial"] & C_{n+1}(X) \arrow[r, "\partial"] \arrow[d, shift right, "f_\#"'] \arrow[d, shift left, "g_\#"] & C_n(X) \arrow[r, "\partial"] \arrow[d, shift right, "f_\#"'] \arrow[d, shift left, "g_\#"] \arrow[dl, "P"'] & C_{n-1}(X) \arrow[r, "\partial"] \arrow[d, shift right, "f_\#"'] \arrow[d, shift left, "g_\#"] \arrow[dl, "P"'] & \cdots \\
\cdots \arrow[r, "\partial"'] & C_{n+1}(X) \arrow[r, "\partial"'] & C_n(Y) \arrow[r, "\partial"'] & C_{n-1}(X) \arrow[r, "\partial"'] & \cdots
\end{tikzcd}$$

2.2.2 Exact Sequences

Let $A \subseteq X$. By $C_n(A)$, we denote the group $C_n(A; \mathbb{Z})$ - the free abelian group generated by continuous functions from Δ^n to A.

It is, of course, a subgroup of $C_n(X)$, and since everything is abelian, all subgroups are normal, so we can construct a chain complex of quotients

$$C_n(X, A) := {C_n(X)}/{C_n(A)}$$

Theorem 2.2.6 (Zigzag Theorem/Long Exact Sequence of Homology)**.** *A short exact sequence* $0 \to A_* \to B_* \to C_* \to 0$ *of chain complexes induces a long exact sequence of homology groups.*

Proof. See [**ICLAlg4**] for a proof that uses the Snake Lemma or [**EPFLRingsAndModules**] for a more fleshed out version. □

Recall the following definition.

Definition 2.2.7. For $A \subseteq X$, we denote by $C_\bullet(X, A)$ the chain complex given by

$$C_n(X, A) = {C_n(X)}/{C_n(A)}$$

for all $n \in \mathbb{N}$.

Theorem 2.2.8 (Long Exact Sequence of a Pair). *For all $A \subseteq X$, there is a LES*

$$\cdots \longrightarrow H_n(A) \longrightarrow H_n(X) \longrightarrow H_n(X, A) \longrightarrow H_{n-1}(A) \longrightarrow \cdots$$

Proof. Exactly Theorem 2.2.6 applied to the SES

$$0 \longrightarrow C_\bullet(A) \longrightarrow C_\bullet(X) \longrightarrow C_\bullet(X, A) \longrightarrow 0$$

□

Say now that we have a space $X = A \cup B$ for some open A, B. We would like to say that $0 \to C_n(A \cap B) \to C_n(A) \oplus C_n(B) \to C_n(X) \to 0$ is short exact, but this is not quite true - we can imagine a simplex S which is only partially contained in each of A, B. However there is an easy fix for this (and it is what you would expect).

Notation. Let X be a space that is covered by open sets A and B. Define

$$C_n^{\{A,B\}}(X)$$

to be the subgroup of $C_n(X)$ generated by those $\sigma : \Delta^n \to X$ such that $\mathrm{im}(\sigma) \subseteq A$ or $\mathrm{im}(\sigma) \subseteq B$.

The idea of the above is that we are somehow "splitting simplices into two parts"—the part contained in A and the part conained in B.

Theorem 2.2.9. *Given a space X and open sets $A, B \subseteq X$ such that $X = A \cup B$, the following is a short exact sequence of chain complexes:*

$$0 \longrightarrow C_*(A \cap B) \longrightarrow C_*(A) \oplus C_*(B) \longrightarrow C_*^{\{A,B\}}(X) \longrightarrow 0$$

Note that in the above SES, if we were to replace $C_*^{\{A,B\}}(X)$ by $C_*(X)$, the sequence would cease to be exact.

Lemma 2.2.10. *For all $n \in \mathbb{N}$, the nth homology group of $C_*(X)$ is the same as the nth homology group of $C_*^{\{A,B\}}(X)$.*

More generally, we show the following.

Lemma 2.2.11. *Let $\mathcal{U} = \{U_j \subseteq X \mid j < J\}$ be subsets of X whose interiors cover X. Define*

$$C_n^{\mathcal{U}}(X) := \left\{ \sum_i \lambda_i \sigma_i \in C_n(X) \;\middle|\; \forall i,\ \exists j \text{ s.t. } \operatorname{im}(\sigma_i) \subseteq U_j \right\}$$

The inclusions $i_* : C_*^{\mathcal{U}}(X) \hookrightarrow C_*(X)$ form a homotopy equivalence of chain complexes.

By the Zigzag theorem, this gives an LES in homology:

Definition 2.2.12 (Mayer-Vietoris Sequence)**.** Given X covered by the interiors of two sets A, B we call the LES

$$\cdots \longrightarrow H_n(A \cap B) \longrightarrow H_n(A) \oplus H_n(B) \longrightarrow H_n(X) \longrightarrow H_{n-1}(A \cap B) \longrightarrow \cdots$$

the **Mayer-Vietoris Sequence for** X.

Say we have a space X and $\mathcal{U} = \{U_j : j \in J\}$ is a cover of X and the interiors of the U_j also cover X. We let $C_n^{\mathcal{U}} \subseteq C_n(X)$ consist of all chains $\sum_{i \in I} \lambda_i \sigma_i$ where for all $i \in I$ we have $\operatorname{im}(\sigma_i) \subseteq U_j$ for some j. Note that the boundary maps ∂ restrict to boundary maps $\partial : C_n\mathcal{U} \to C_{n-1}^{\mathcal{U}}(X)$.

We now temporarily call the corresponding homology groups $H_n^{\mathcal{U}}(X)$ ("temporarily" because we will hopefully soon show that $H_n^{\mathcal{U}}(X)$ is just $H_n(X)$ for every $\mathcal{U}$):

Lemma 2.2.13. *The inclusion map $i : C_n^{\mathcal{U}}(X) \to C_n(X)$ is a chain homotopy equivalence (i.e. the projection $p : C_n(X) \to C_n^{\mathcal{U}}(X)$ is such that $p \circ i$ and $i \circ p$ are chain homotopic to the identity), and in particular we have $H_n^{\mathcal{U}}(X) \cong H_n(X)$.*

Proof. We just split Δ^n into many smaller simplices, for example by repeated barycentric subdivisions... we then find a homotopy of chain complexes between $C_*(X)$ and $C_*^{\{A,B\}}(X)$.

Let $Y \subseteq \mathbb{R}^d$ be a convex set, and let $LC_n(Y) \subseteq C_n(Y)$ denote the subgroup of affine maps $\sigma : \Delta^n \to Y$. The boundary map ∂ restricts to a map $\partial : LC_n(Y) \to LC_{n-1}(Y)$. For every $b \in Y$, define

$$b : LC_n(Y) \to LC_{n+1}(Y) : [w_0, \ldots, w_n] \mapsto [b, w_0, \ldots, w_n]$$

It is not difficult to show that

$$\partial \circ b = \mathrm{id} - b \circ \partial \iff \partial \circ b + b \circ \partial = \mathrm{id}$$

We now let $S : LC_n(Y) \to LC_n(Y)$ be defined inductively by $S : LC_0(Y) \to LC_0(Y)$ is the identity, and $S_\sigma = b_\sigma(S\partial\sigma)$, where B_σ is the barycenter of $S\partial\sigma$. Then $\partial S_\sigma = \partial b_\sigma(S\partial\sigma) = S\partial\sigma - b_\sigma\partial$sorry □

2.2.3 Application: Homology Groups of Spheres

Let B_n denote the closed unit ball in $\mathbb{R}^n$ centred at the origin, so that $S^{n-1} = \partial B_n \subseteq B_n$. Then, the following is a long exact sequence:

$$\cdots \longrightarrow H_k(S^{n-1}) \longrightarrow H_k(B_n) \longrightarrow H_k(B_n, S^{n-1}) \longrightarrow H_{k-1}(S^{n-1}) \longrightarrow \cdots$$

We can show the following fact.

Theorem 2.2.14. *$H_n(S^n; \mathbb{Z}) \cong \mathbb{Z}$ for all $n \geq 1$.*

Proof. □

2.2.4 Excision

(Recall that relative homology is obtained by quotienting out chain complexes in X by chain complexes in A and then computing homology.)

Theorem 2.2.15 (Excision). *Let $A, B \subseteq X$, with the interiors covering X. The inclusion*

of $(B, A \cap B)$ *into* (X, A) *induces isomorphisms*

$$H_n(B, A \cap B) \xrightarrow{\sim} H_n(X, A)$$

for all $n \in \mathbb{N}$. *Equivalently, suppose we have a chain of inclusions* $Z \subseteq A \subseteq X$, *with* $\operatorname{cl} Z \subseteq \operatorname{int}(A)$. *Then,*

$$H_n(X, A) \cong H_n(X \setminus Z, A \setminus Z)$$

These equivalent phenomena are called ***excision***.

Proof. We have a diagram

$$C_n(B)\big/C_n(A \cap B) \longrightarrow C_n^{\{A,B\}}(X)\big/C_n(A) \xrightarrow{\text{induces isomorphism on } H_*} C_n(X)\big/C_n(A)$$

□

We can also talk about nested inclusions.

Proposition 2.2.16 (Long Exact Sequence of a Triple). *Given inclusions* $A \subseteq B \subseteq X$, *the following is a short exact sequence of chain complexes:*

$$0 \longrightarrow C_\bullet(B, A) \longrightarrow C_\bullet(X, A) \longrightarrow C_\bullet(X, B) \longrightarrow 0$$

This again induces a long exact sequence of homology.

Example 2.2.17 ($\star \subseteq S^1 \subseteq D^2$). Consider a point $\star \in S^1$ and consider the inclusions given by $\{\star\} \subseteq S^1 \subseteq D^2$. We get a long exact sequence

$$\cdots \longrightarrow H_2(S^1, \star) \longrightarrow H_2(D^2, \star) \longrightarrow H_2(D^2, S^1) \longrightarrow H_1(S^1, \star) \longrightarrow H_1(D^2, \star) \longrightarrow H$$

We can use what we know about long exact sequences of a pair to compute $H_n(S^1, \star)$:

$$\cdots \longrightarrow \underbrace{H_n(\star)}_{=0} \longrightarrow H_n(S^1) \xrightarrow{\sim} H_n(S^1, \star) \longrightarrow \underbrace{H_{n-1}(\star)}_{=0} \longrightarrow \cdots$$

In particular, $H_n(S^1, \star) \cong \tilde{H}_n(S^1, \star)$. (And the same is true of other spaces as well, not

just S^1.)

2.2.5 Changing the Base Group

Suppose I have a short exact sequence of abelian groups

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

It turns out this induces the following long exact sequence of chain complexes:

$$0 \longrightarrow C_\bullet(X; A') \longrightarrow C_\bullet(X; A) \longrightarrow C_\bullet(X; A'') \longrightarrow 0$$

where at every stage, we compute the cycle groups with respect to a different base group.

2.2.6 Good Pair

Definition 2.2.18 (Good Pair). A pair (X, A) of some $A \subseteq X$ is a **good pair** if A is nonempty and closed and there exists a neighbourhood V of A in X that deformation retracts to A.

2.2.7 Reconciling Singular and Simplicial Homology

Recall the Five Lemma.

Lemma 2.2.19 (Five Lemma). *Consider the following diagram in some abelian category, with the top and bottom rows both being exact:*

$$\begin{tikzcd}
A \arrow[r, "i"] \arrow[d, dashed, "\alpha"] & B \arrow[r, "j"] \arrow[d, dashed, "\beta"] & C \arrow[r, "k"] \arrow[d, dashed, "\gamma"] & D \arrow[r, "l"] \arrow[d, dashed, "\delta"] & E \arrow[d, dashed, "\epsilon"] \\
A' \arrow[r, "i'"] & B' \arrow[r, "j'"] & C' \arrow[r, "k'"] & D' \arrow[r, "l'"] & E'
\end{tikzcd}$$

The following are all true.

1. *If β, δ are surjective and ϵ is injective, then γ is surjective.*
2. *If β, δ are injective and α is surjective, then γ is injective.*

In particular, if $\alpha, \beta\, \delta, \epsilon$ are isomorphisms, then γ is also an isomorphism.

Proof. We begin the grand diagram chase. Note: we can also prove the first two as Four Lemmas - see [**EPFLRingsAndModules**]. For a version of this statement, see [**EPFLRingsAndModules**].

1. Fix $c' \in C'$. There is $d \in D$ such that $\delta(d) = k'(c')$. Notice that $\epsilon(l(d)) = l'(\delta(d)) = l'(k'(c')) = 0$. Since ϵ is injective, this tells us that $l(d) = 0$. Thus, $d = k(c)$ for some $c \in C$. Then,

$$\begin{aligned} k'(c' - \gamma(c)) &= k'(c') - k'(\gamma(c)) \\ &= k'(c') - \delta(k(c)) \\ &= k'(c') - \delta(d) && = 0 \end{aligned}$$

   sorry

2. Fix $c \in C$ with $\gamma(c)$

The union of the first two gives you immediately that if α, β, δ and ϵ are isomorphisms, then so is γ. □

We are now ready to show that singular and simplicial homology are isomorphic. In fact, they are naturally isomorphic as functors.

Theorem 2.2.20. *Let X be a Δ-complex and let $A \subseteq X$ be a subcomplex. Then,*

$$\underbrace{H_\bullet^\Delta(X, A)}_{\textit{simplicial}} \cong \underbrace{H_\bullet(X, A)}_{\textit{singular}}$$

naturally in X and A.

Proof. □

Example 2.2.21 (A Weird Pathology)**.** Define a Δ-complex by gluing together countably many copies of the unit interval at the same endpoint. So it looks like an asterisk with countably many spokes (the caveat being this space is actually not metrisable, and in particular, doesn't embed into any Euclidean space, let alone the plane). A path that goes halfway along the first spoke before going back to the vertex, a quarter of the way along

the second before going back to the vertex, an eight of the way along the third before going back, and so on.

Chapter 3

Cohomology

In this chapter, we will study the dual notion of homology: cohomology.

3.1 Some Motivation

3.1.1 Degrees of Maps on n-Spheres

Definition 3.1.1 (Degree)**.** Let f be a continuous map from S^n to itself. We know there is an induced map $f_* : H_n(S^n) \to H_n(S^n)$. We know that $H_n(S^n) \cong \mathbb{Z}$. Factoring through this isomorphism, we call $f_*(1)$ the **degree** of f, denoted $\deg(f)$.

The intuition behind this number is that it tells you "how many times your map f wraps the n-sphere around itself."

Lemma 3.1.2 (Facts About the Degree)**.**

1. *If f is not surjective then* $\deg(f) = 0$.
2. $\deg(f \circ g) = \deg(f) \cdot \deg(g)$.
3. $\deg(\mathrm{id}) = 1$.
4. *If $f \simeq g$ (ie, f is homotopic to g), then* $\deg(f) = \deg(g)$.[a]
5. *If f is a reflection of some map via a hyperplane, then* $\deg(f) = -1$.

[a] The converse is *also* true: it's called Hopf's Theorem.

Proof Sketch.

1. If f is not surjective, then there is some $x_0 \in S^n$ such that f corestricted to $S^n \setminus \{x_0\}$ agrees with f. That is, the following diagram commutes:

$$\begin{tikzcd} S^n \arrow[rr, "f"] \arrow[dr, "f"'] & & S^n \\ & S^n \setminus \{x_0\} \cong B^n \arrow[ur, hook] & \end{tikzcd}$$

   By functoriality of homology, we know that taking H_n preserves the above commutative diagram. sorry

2. This follows from the functoriality of homology.

3. This also follows from the functoriality of homology.

4. Something to do with agreement of homology on homotopic spaces/maps? sorry

5. sorry

□

To see why this is useful, we consider groups acting on n-spheres.

Recall the notion of a **free action**.

Definition 3.1.3 (Free Group Action)**.** A group G **acts freely** on a set X if the action has no fixed points, ie, if every nontrivial element of G fixes no elements of X.

It turns out maps with no fixed points have a special property.

Lemma 3.1.4. *If $f : S^n \to S^n$ is continuous and has no fixed points, then f is homotopy equivalent to the antipodal map $a : S^n \to S^n : x \mapsto -x$. In particular,* $\deg(f) = (-1)^{n+1}$.

Proof Sketch. We can define an explicit homotopy $F : S^n \times I \to S^n$ by

$$F(x, t) = \frac{t \cdot f(x) + (1 - t) \cdot (-x)}{\|t \cdot f(x) + (1 - t) \cdot (-x)\|}$$

It is not difficult to show that this is a well-defined homotopy. □

We can now use the machinery about degrees to show the following rather nontrivial fact.

Theorem 3.1.5. *Let n be an even number. The only nontrivial group that acts freely on S^n is $\mathbb{Z}/2\mathbb{Z}$.*

Proof. Let G be a group acting freely on S^n. Then,

$$\deg : G \to \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$$

is a group homomorphism.

Since the action is free, every $g \in G \setminus \{\mathrm{id}\}$ is a homeomorphism without fixed points. sorry □

Note that the above is really not true of odd-dimensional spheres.

Counterexample 3.1.6 (Many Free Actions on S^3)**.** We can view S^3 as a subset of $\mathbb{C}^2$. We can show that $\mathbb{Z}/n\mathbb{Z}$, generated by some ϵ, acts freely on S^3. Indeed, map any pair $(z, w) \in \mathbb{C}^2$ to the point

$$\left(e^{\frac{2\pi i}{n}} \cdot z, e^{\frac{2\pi i}{n}} \cdot w\right)$$

In other words, we rotate both coordinates in some special way. This action has no fixed points when restricted to S^3. Fun fact: the quotient of S^3 by the above action is called a **lens space**.

We can do something similar for any odd-dimensional sphere, as it lives inside a Euclidean space of even dimension.

Theorem 3.1.7. *S^n has a nonzero tangent vector field (that is, $\{v : S^n \to \mathbb{R}^{n+1} \mid \langle v(x), x\rangle = 0\} \neq 0$) if and only if n is odd.*

Proof. sorry □

Finally, we outline some useful homology theoretic machinery for computing degrees.

It turns out the following is a short exact sequence of chain complexes:

$$0 \longrightarrow C_\bullet(\mathbb{R}P^n; \mathbb{Z}/2\mathbb{Z}) \xrightarrow{\tau} C_\bullet(S^n; \mathbb{Z}/2\mathbb{Z}) \xrightarrow{q_\bullet} C_\bullet(\mathbb{R}P^n; \mathbb{Z}/2\mathbb{Z}) \longrightarrow 0$$

where $q : S^n \to \mathbb{R}^n$ is given by mapping σ to $\tilde{\sigma_1} + \tilde{\sigma_2}$, where $\tilde{\sigma_1}$ and $\tilde{\sigma_2}$ are the two lifts of σ.

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